

STOCKHOLM VASTERAS, Sweden

WMO: 024460

Lat: 59.589N

Lon: 16.634E

Elev: 6

StdP: 101.25

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.1	-15.0	-20.9	0.6	-18.0	-17.8	0.8	-14.2	11.3	2.7	10.2	2.4	0.9	0	0.618

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	27.2	18.2	25.8	17.5	23.9	16.7	19.6	25.1	18.6	23.7	17.6	22.3	3.7	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.8	12.8	21.4	16.8	12.0	20.6	15.8	11.2	20.0	55.7	25.0	52.5	23.7	49.6	22.2	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.3	8.3	7.4	-21.5	29.6	4.7	1.8	-24.9	30.9	-27.6	32.0	-30.2	33.0	-33.6	34.3		
			-21.8	20.9	4.6	1.4	-25.1	21.9	-27.8	22.7	-30.4	23.4	-33.7	24.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.6	-3.2	-3.1	0.4	5.6	10.5	14.8	17.5	16.5	11.9	7.0	2.2	-1.0
	DBStd	8.32	5.42	5.39	4.27	3.41	3.90	2.93	2.75	2.77	2.99	3.43	4.03	5.57
	HDD10.0	1954	411	367	299	139	42	2	0	1	16	105	233	341
	HDD18.3	4308	669	601	557	383	245	110	49	69	192	352	483	599
	CDD10.0	730	0	0	0	6	56	147	233	202	74	11	1	0
	CDD18.3	43	0	0	0	0	1	6	24	13	1	0	0	0
	CDH23.3	482	0	0	0	0	22	77	263	119	1	0	0	0
	CDH26.7	82	0	0	0	0	1	7	58	16	0	0	0	0
	WSAvg	3.5	3.7	3.5	3.6	3.5	3.6	3.4	3.2	3.1	3.4	3.6	3.7	3.7
	PrecAvg	544	34	25	27	31	36	48	74	70	59	51	50	39
Precipitation	PrecMax	798	64	57	57	72	93	111	151	154	192	110	96	116
	PrecMin	399	4	6	0	1	9	5	24	8	16	11	19	10
	PrecStd	94	16	14	15	17	24	26	35	37	37	31	22	22
	PrecStd	94	16	14	15	17	24	26	35	37	37	31	22	22
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.1	8.1	13.8	20.2	25.8	27.9	30.9	28.7	23.0	16.2	12.0	10.2
		MCWB	5.9	5.2	8.3	11.4	16.4	17.5	19.9	18.4	16.1	13.7	10.4	8.4
	2%	DB	5.9	6.1	10.2	17.8	23.2	25.8	28.2	26.2	20.9	14.8	10.2	8.2
		MCWB	4.7	4.2	5.9	10.1	14.6	16.8	19.0	18.1	15.7	12.5	9.1	6.8
	5%	DB	4.8	4.9	8.0	14.9	21.0	23.2	26.2	24.4	18.9	13.2	9.1	6.9
		MCWB	3.8	3.4	4.8	9.0	13.6	15.7	18.0	17.3	14.5	11.4	8.3	5.6
	10%	DB	3.2	3.8	6.1	12.2	18.1	21.2	24.2	22.8	17.2	12.2	8.0	5.2
		MCWB	2.2	2.5	3.6	7.1	11.9	14.8	17.4	16.5	13.7	10.4	7.2	4.3
	0.4%	WB	6.0	5.9	8.2	12.6	17.7	19.2	21.3	20.6	17.3	14.4	10.8	8.9
		MCDB	6.5	7.0	12.1	18.9	23.7	24.7	26.8	25.5	20.6	15.8	11.7	10.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	4.6	4.4	6.4	10.6	15.7	17.7	20.2	19.2	16.1	13.1	9.5	7.0
		MCDB	5.2	5.5	9.3	15.9	20.9	23.5	26.2	23.9	19.4	14.3	10.2	7.9
	5%	WB	3.6	3.2	5.2	9.4	14.1	16.6	19.1	18.2	15.1	12.1	8.1	5.5
		MCDB	4.3	4.2	7.2	14.0	19.9	21.9	24.8	22.5	18.0	13.4	8.8	6.3
	10%	WB	2.3	2.4	3.9	7.9	12.7	15.5	18.1	17.3	14.1	11.0	7.1	4.3
		MCDB	3.1	3.3	5.7	11.4	17.6	20.4	23.2	21.3	16.7	12.2	7.6	5.0
Mean Daily Temperature Range	5% DB	MDBR	5.3	6.4	7.6	10.3	10.8	10.1	10.4	10.2	8.9	6.5	4.6	4.9
		MCDBR	4.6	5.9	9.9	14.8	15.5	14.3	14.5	13.4	11.4	7.1	5.3	5.2
		MCWBR	4.5	4.6	6.7	8.7	8.5	7.3	6.8	6.5	6.9	5.5	4.9	4.8
	5% WB	MCDBR	4.5	5.2	8.8	13.3	13.6	12.1	12.8	11.2	10.0	6.3	5.2	4.8
		MCWBR	4.5	4.4	6.3	8.2	7.9	6.8	6.7	5.9	6.4	5.2	4.9	4.6
Clear-Sky Solar Irradiance	taub		0.274	0.316	0.338	0.374	0.371	0.361	0.386	0.380	0.344	0.332	0.301	0.257
	taud		2.203	2.270	2.352	2.319	2.392	2.441	2.392	2.406	2.492	2.424	2.263	2.131
	Ebn at Noon		517	687	796	825	858	872	837	813	789	670	478	385
	Edh at Noon		43	69	86	106	106	103	106	97	76	60	41	31
All-Sky Solar Radiation	RadAvg		0.31	0.95	2.46	3.93	5.08	5.68	5.21	4.12	2.70	1.21	0.39	0.20
	RadStd		0.05	0.14	0.31	0.39	0.47	0.36	0.52	0.41	0.31	0.20	0.09	0.02

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (58 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	-121	-147	N/A	N/A

Nomenclature: See separate page